TECHNICAL

Technical Interview Questions (Core CS)

Operating Systems & Database Management

1. Operating Systems High-Yield Topics

- Process vs. Thread: Process is an execution program with dedicated address space; Thread is a lightweight sub-unit sharing process memory.
- Deadlock Conditions: Mutual exclusion, Hold & wait, No preemption, Circular wait.
- Virtual Memory & Paging: Eliminates external fragmentation; Page replacement algorithms: FIFO, LRU, LFU.
- Context Switching: Overhead of saving state of currently running process and restoring next scheduled process.

2. DBMS & SQL Query Architecture

- ACID Properties: Atomicity (all or nothing), Consistency, Isolation, Durability.
- Normalization: 1NF (atomic values), 2NF (no partial dependency), 3NF (no transitive dependency), BCNF.
- Clustered vs. Non-Clustered Index: Clustered alters physical row storage order (one per table); Non-clustered creates index table with row pointers.
- Transactions & Locking: Shared locks (read-only) vs. Exclusive locks (write access).

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Computer Networks & System Design Principles

3. Computer Networks (OSI & TCP/IP)

- OSI 7 Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
- TCP vs. UDP: TCP is connection-oriented, reliable, with flow & congestion control. UDP is stateless, lightweight, low-latency.
- DNS Resolution: Browser cache -> OS cache -> Router cache -> ISP DNS Resolver -> Root/TLD servers.
- HTTPS & TLS Handshake: Symmetric key encryption negotiated via asymmetric RSA/ECC key exchange.

4. System Design Basics for Freshers

- Horizontal Scaling vs. Vertical Scaling: Adding more servers vs. upgrading CPU/RAM of a single machine.
- Load Balancing: Distributes traffic (Round Robin, Least Connections, IP Hash).
- Caching Strategies: Write-Through, Write-Back, Cache-Aside using Redis / Memcached.
- CAP Theorem: A distributed system can guarantee at most 2 out of 3: Consistency, Availability, Partition Tolerance.